# 8D 1 MAKING STANDARD SOLUTIONS

SPECIFICATION REFERENCE
8.20 PART CP4

## LEARNING OBJECTIVES

- Be able to calculate solution concentrations, including simple acid-base titrations.

### WHAT ARE STANDARD SOLUTIONS AND PRIMARY STANDARDS?

The next topic, **Topic 8D.2**, is about doing titrations. One substance needed in a titration is a standard solution, which we will look at in this topic.

A **standard solution** is a solution whose concentration is accurately known. One obvious way to prepare a standard solution is to take a known mass of a substance and dissolve it in water to make a known volume of solution.

These substances are known as **primary standards**. Ideally, primary standards should:

- be solids with high molar masses
- be available in a high degree of purity
- be chemically stable (neither decompose nor react with substances in the air)
- not absorb water from the atmosphere
- be soluble in water
- react rapidly and completely with other substances when used in titrations.

Unfortunately, several substances that are often used in titrations are not suitable as primary standards. For example, hydrochloric acid does not exist as a solid, but only as HCl(g) and HCl(aq). Sodium hydroxide is a solid, but it readily absorbs water vapour and reacts with carbon dioxide from the atmosphere. A sample of sodium hydroxide can be accurately weighed, but it is not certain what is being weighed, as there will be unknown masses of water and sodium carbonate mixed with it. You may remember seeing a white powder around the neck of a bottle of sodium hydroxide solution. This white powder is sodium carbonate.

### MAKING A STANDARD SOLUTION OF SULFAMIC ACID

Sulfamic acid is probably unfamiliar to you. It is a readily available primary standard for use in acid–base titrations that has the necessary characteristics. Its formula can be shown in more than one way, including $NH_2SO_3H$, and it has a molar mass of $97.1\,g\,mol^{-1}$.

We will look in detail at a method used to obtain an accurately known value for the concentration of this solution. The method used is known as 'weighing by difference'. In your practical work, you may use a different method.

### CALCULATING ROUGHLY HOW MUCH TO WEIGH

You need some idea of the approximate concentration and volume of the solution to be made. Typical values are $0.1\,mol\,dm^{-3}$ and $250\,cm^3$.

Using the calculation method given in **Topic 1E.2**, the approximate amount of sulfamic acid, $n = c \times V = 0.1 \times 0.25 = 0.025\,mol$, so the approximate mass needed is $m = n \times M = 0.025 \times 97.1 = 2\,g$.

Note that even though we are going to do a very accurate weighing, we only need to know an approximate mass at this stage.

### APPARATUS

The apparatus you need is:

- safety glasses and a lab coat
- an accurate balance (we will assume one reading to 3 decimal places)
- a weighing bottle (or weighing boat)
- a spatula
- a $250\,cm^3$ beaker
- a $250\,cm^3$ volumetric flask
- a wash bottle containing deionised water (or distilled water)
- a small funnel
- a glass stirring rod.

### METHOD

1 Add between 2.3 and 2.5 g of sulfamic acid to the weighing bottle and weigh accurately.
2 Transfer as much as possible of the acid to a clean beaker and reweigh the weighing bottle.
3 Add about $100\,cm^3$ of deionised water to the beaker and stir until all of the sulfamic acid has dissolved.
4 Remove the stirring rod, washing traces of solution from the rod into the beaker using the wash bottle.
5 Place a funnel in the neck of the volumetric flask and pour the solution from the beaker into the flask.
6 Rinse the inside of the beaker several times using the wash bottle and transfer the rinsings to the flask.
7 Add deionised water to the flask and fill up exactly to the graduation mark.
8 Stopper the flask and invert it several times to make a uniform solution.

You can now calculate an accurate value for the concentration of the solution, using these example values.

mass of weighing bottle + sulfamic acid = 19.542 g

mass of weighing bottle + any traces of sulfamic acid = 17.151 g

mass of sulfamic acid added = 2.391 g

$$n(NH_2SO_3H) = \frac{2.391}{97.1} = 0.02462\,mol$$

$$c = \frac{0.02462}{0.250} = 0.0985\,mol\,dm^{-3}$$

## LEARNING TIP

Look carefully at each step of the method. How does each step contribute to the accuracy of the final value of the concentration?







When making a standard solution in a volumetric flask, it is important to ensure that the lowest part of the liquid (the meniscus) is aligned with the graduation mark.

## CHECKPOINT

**1.** A student used the method described but made two mistakes.

(a) In Step 6, he poured the rinsings down the sink instead of transferring them to the flask.

(b) In Step 7, he added water above the graduation mark.

Explain how each mistake affected the calculated concentration.

SKILLS CRITICAL THINKING

**2.** A student used the method to make 500 $cm^3$ of a solution of sodium carbonate (M = 106.0 $g\,mol^{-1}$). These are her weighings:

mass of weighing bottle + sodium carbonate = 23.382 g

mass of weighing bottle = 18.218 g

Calculate the concentration of the solution made.

SKILLS PROBLEM SOLVING

## SUBJECT VOCABULARY

**standard solution** a solution whose concentration is accurately known

**primary standards** substances used to make a standard solution by weighing

# 8D 2 DOING TITRATIONS

SPECIFICATION REFERENCE

8.20 PART | CP3 | CP4

## LEARNING OBJECTIVES

- Be able to calculate solution concentrations, including simple acid-base titrations.

## WHAT IS A TITRATION?

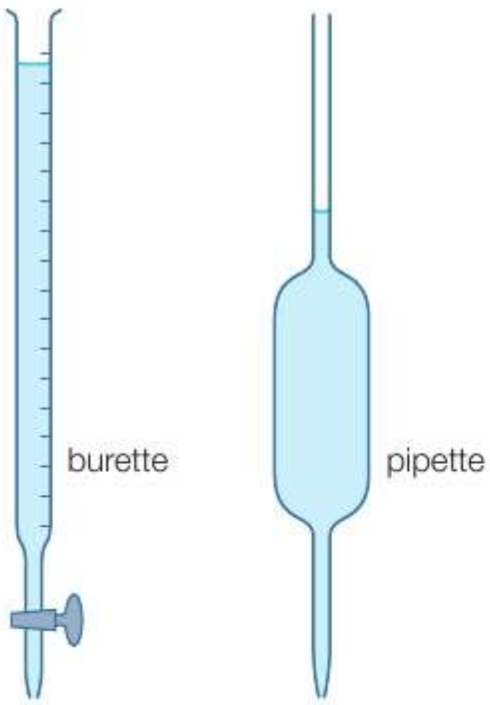
A titration is a practical method with the aim of measuring the volumes of two solutions that react together, and using the results to calculate the concentration of one of the solutions. Because the method involves measuring volumes, it is sometimes known as volumetric analysis.

You may come across different titration types (redox titrations and complexometric titrations), but in this book we will consider only acid–base titrations. Many, but not all, bases are soluble in water, which makes them alkalis. They are often referred to as acid–alkali titrations.

All of the acids and bases you will use in titrations are colourless, as are the products, so there is nothing to see when the reaction is complete. This problem is solved by using an indicator, a substance which has different colours in acids and bases/alkalis.

In the method described in this topic, we will assume that a solution of sodium hydroxide of approximate concentration 0.1 mol $dm^{-3}$ is available and that the standard solution of sulfamic acid referred to in **Topic 8D.1** will be used. The aim is to calculate an accurate value for the concentration of the sodium hydroxide solution. The method will be described in some detail, and we will focus on avoiding common errors in technique.

In some titrations, it is important to choose which way round to do the titration (which solution goes in the pipette and which solution in the burette). In other titrations, it does not matter.




▲ **fig A** Note that the space between the tap and the tip of the burette is full of solution.

## OUTLINE OF THE TITRATION METHOD

Here is an outline of the titration method that introduces some key terms.

- Add the acid to the alkali until the **equivalence point** of the titration and the **end point** of the indicator is reached.
- Record the burette readings using the lowest part of the **meniscus**.
- Calculate the **titre**.
- Repeat the titration until **concordant titres** are obtained.

## APPARATUS

The apparatus likely to be used is:

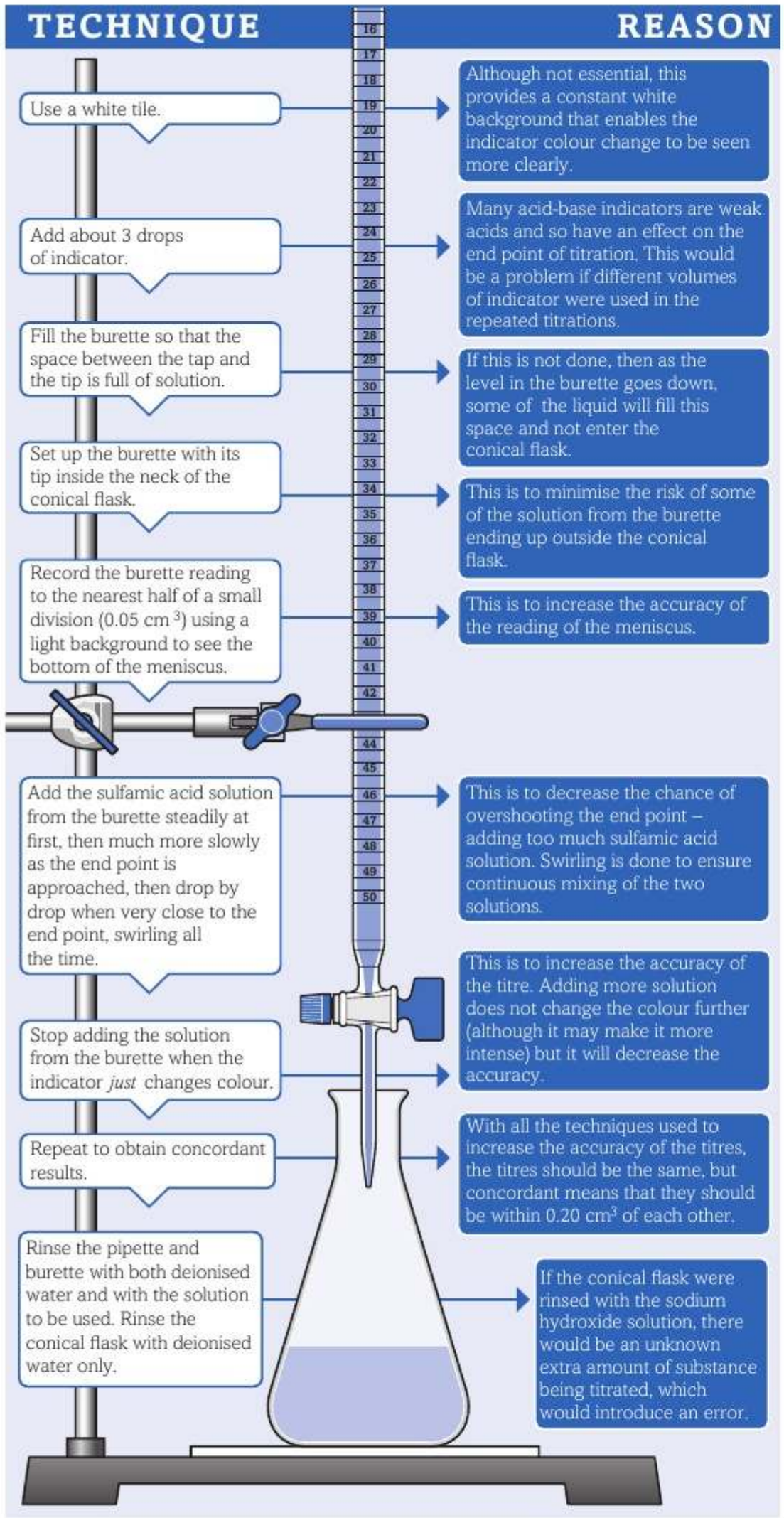
- a conical flask (usually 250 $cm^3$)
- a burette (usually 50 $cm^3$) and stand
- a pipette (usually 25 $cm^3$) and pipette filler
- a wash bottle containing deionised water (or distilled water)
- a small funnel
- a white tile.

## METHOD

1. Rinse the conical flask with deionised water and place it on a white tile.
2. Using a pipette filler, rinse the pipette with deionised water and then with some of the sodium hydroxide solution.
3. Use the pipette to transfer 25.0 $cm^3$ of the sodium hydroxide solution to the conical flask.
4. Add about 3 drops of methyl orange indicator.
5. Rinse the burette with deionised water and then with some of the sulfamic acid solution.
6. Fill the burette with the sulfamic acid solution and set it up in the stand above the conical flask.
7. Record the burette reading.
8. Add the sulfamic acid solution to the conical flask until the indicator just changes colour, and again record the burette reading.
9. Empty and rinse the conical flask with deionised water, and repeat the titration until concordant titres have been obtained.

## TITRATION TECHNIQUES

To obtain accurate results in a titration, it is important to work carefully. The diagram shows some important techniques and the reasons for them.




▲ **fig B** Titration techniques and the reasons for them.

## CHOOSING AN INDICATOR

Two common indicators are:

- methyl orange
- phenolphthalein.

Sometimes it is important to use one of these and not the other, but in other titrations it does not matter which one you use.

**Table A** shows the colours of these indicators and which combination of acid and base they should be used with.

| INDICATOR | COLOUR IN ACID | COLOUR IN ALKALI | ACID–BASE COMBINATION |
|---|---|---|---|
| methyl orange | red | yellow | strong acid – weak base and strong acid – strong base |
| phenolphthalein | colourless | pink | weak acid – strong base and strong acid – strong base |

**table A** Information about methyl orange and phenolphthalein.

Examples of strong acids are:

- hydrochloric acid
- nitric acid.

Examples of strong bases are:

- sodium hydroxide
- potassium hydroxide.

The commonest weak base is ammonia.

The commonest weak acid is ethanoic acid.



▲ **fig C** This shows the burette after the titration has been done. Note the white tile and the colour of the phenolphthalein indicator. The pink colour shows that an acid has been neutralised and there is an excess of an alkali.

### LEARNING TIP

The best way to understand all the features of successful titrations is to do several titrations yourself.

### CHECKPOINT

SKILLS CRITICAL THINKING

1. A student does not fill the burette space between the tap and the tip in a titration. Explain the effect of this mistake on the value of the titre.

2. A student rinses out the conical flask with deionised water, then with the solution used in the pipette. Explain the effect of this mistake on the value of the titre.

### SUBJECT VOCABULARY

**equivalence point** the point at which there are exactly the right amounts of substances to complete the reaction

**end point** the point at which the indicator just changes colour; ideally, the end point should coincide with the equivalence point

**meniscus** the curving of the upper surface of a liquid in a container; the lowest (horizontal) part of the meniscus should be read

**titre** the volume added from the burette during a titration

**concordant titres** titres that are close together (usually within 0.20 $cm^3$ of each other)

# 8D 3 CALCULATIONS FROM TITRATIONS

SPECIFICATION REFERENCE

8.20 PART | CP3 | CP4

## LEARNING OBJECTIVES

- Be able to calculate solution concentrations, including simple acid-base titrations.

### CALCULATING THE MEAN (AVERAGE) TITRE

The terms 'average' and 'mean' are often used interchangeably, but in scientific work you should take care to use the proper word. 'Average' has more than one mathematical meaning – it can represent the mean, the median or the mode. For these calculations you should use the mean: a set of values added to give a total, which is then divided by the number of values.

Before calculating a mean titre, only the concordant values must be selected, i.e. those that are within 0.20 $cm^3$ of each other. **Table A** shows some typical titration results and a student's choice of concordant titres.

| TITRATION NUMBER | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| final burette reading / $cm^3$ | 24.15 | 25.30 | 24.60 | 23.25 |
| initial burette reading / $cm^3$ | 1.20 | 2.70 | 1.90 | 0.60 |
| titre / $cm^3$ | 22.95 | 22.60 | 22.70 | 22.65 |
| concordant titres | ✘ | ✔ | ✔ | ✔ |

**table A** A set of titration results and a student's choice of concordant titres.

In this example, titrations 2, 3 and 4 are all within 0.20 $cm^3$ of each other and so have been correctly ticked as concordant. It is not surprising that titration 1 is not concordant because it is normal practice to do the first titration more quickly to obtain a rough titre, so the end point is more likely to be overshot. This saves time in the long run because in the other titrations the liquid in the burette can be added quickly at first, until the end point is close, then added much more slowly.

This is how to calculate the mean:

$$\text{Mean} = \frac{22.60 + 22.70 + 22.65}{3} = 22.65 \text{ cm}^3$$

### CALCULATING A CONCENTRATION

#### WORKED EXAMPLE 1

Here is a typical set of titration results from the method given in **Topic 8D.2**.

volume of sodium hydroxide solution used = 25.0 $cm^3$

volume of sulfamic acid solution used = 22.65 $cm^3$

concentration of sulfamic acid solution = 0.0985 mol $dm^{-3}$

The equation for the reaction is:

$$NaOH(aq) + NH_2SO_3H(aq) \rightarrow NH_2SO_3Na(aq) + H_2O(l)$$

You may be shown a shorter method than the following one, but this method, using moles, is more versatile and can be used in more difficult examples.

Step 1: calculate the amount of one substance, in this case the sulfamic acid.

$$n(NH_2SO_3H) = c \times V = 0.0985 \times \frac{22.60}{1000} = 0.00223 \text{ mol}$$

Step 2: calculate the amount of the other substance using the reacting ratio in the equation.

As the reacting ratio is 1 : 1, $n(NaOH) = 0.00223$ mol

Step 3: calculate the concentration of sodium hydroxide solution.

$$c = \frac{n}{V} = \frac{0.00223}{0.0250} = 0.0892 \text{ mol dm}^{-3}$$

## WORKED EXAMPLE 2

A titration is done to calculate the concentration of a solution of nitric acid, using a standard solution of sodium carbonate. The equation for the reaction is:

$$Na_2CO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + CO_2$$

The titration results are:

volume of sodium carbonate solution used = $25.0\,cm^3$

volume of nitric acid solution used = $27.25\,cm^3$

concentration of sodium carbonate solution = $0.108\,mol\,dm^{-3}$

Step 1: calculate the amount of sodium carbonate.

$$n = 0.108 \times \frac{25}{1000} = 0.00270\,mol$$

Step 2: calculate the amount of the other substance using the reacting ratio in the equation (1 : 2 in this example).

$$n = 0.00270 \times 2 = 0.00540\,mol$$

Step 3: calculate the concentration of the nitric acid.

$$c = \frac{0.00540}{0.02725} = 0.198\,mol\,dm^{-3}$$

## WORKED EXAMPLE 3

A titration is done to calculate the concentration of a solution of hydrochloric acid, using the sodium hydroxide solution from Worked example 1. The equation for the reaction is:

$$NaOH + HCl \rightarrow NaCl + H_2O$$

The titration results are:

volume of sodium hydroxide solution used = $25.0\,cm^3$

volume of hydrochloric acid used = $22.68\,cm^3$

concentration of sodium hydroxide solution = $0.0892\,mol\,dm^{-3}$

This time, we will use the shorter method of calculation, using the expression

$$V_1 \times \frac{M_1}{n_1} = V_2 \times \frac{M_2}{n_2}$$

where $V$ is the volume, $M$ is the molar concentration and $n$ is the coefficient in the equation. The subscript 1 refers to the first substance in the equation, and the subscript 2 to the second substance.

In this example, the unknown is $M_2$, so rearranging the expression and substituting the values gives the answer.

$$M_2 = \frac{V_1 \times M_1 \times n_2}{V_2 \times n_1} = \frac{25.0 \times 0.0892 \times 1}{22.68 \times 1} = 0.0983\,mol\,dm^{-3}$$

**LEARNING TIP**

Practise the calculation method. Make sure that you use the reacting ratio the right way round.

## CHECKPOINT

**1.** A student records these readings during a titration:

| | | | | |
|---|---|---|---|---|
| final burette reading / $cm^3$ | 28.0 | 27.8 | 27.4 | 27.0 |
| initial burette reading / $cm^3$ | 2.7 | 2.8 | 2.8 | 2.8 |
| titre / $cm^3$ | 25.3 | 25.0 | 24.6 | 24.2 |

mean titre = $24.4\,cm^3$

What mistakes has the student made in recording these results?

SKILLS CRITICAL THINKING

**2.** A student does a titration using this reaction:

$$2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$$

She records these results:

volume of KOH solution = $25.0\,cm^3$

volume of $H_2SO_4$ = $19.83\,cm^3$

concentration of $H_2SO_4$ = $0.0618\,mol\,dm^{-3}$

What is the concentration of the KOH solution, in $mol\,dm^{-3}$?

SKILLS PROBLEM SOLVING

# 8D 4 MISTAKES, ERRORS, ACCURACY AND PRECISION

SPECIFICATION REFERENCE 8.22

## LEARNING OBJECTIVES

- Understand the difference between a mistake and an error.
- Understand the difference between accuracy and precision.
- Comment on sources of error in experimental procedures.

## USING THE CORRECT TERMINOLOGY

In science, it is sometimes difficult to find the correct words to use when considering the results of experiments and the calculations based on these results. This is because in the non-scientific world, words are often used with less care. You have already seen the idea that 'amount' has a specific meaning in chemistry (amount of substance, in moles) and should not be used to refer to mass or volume.

We will look at terminology in this topic.

- Mistakes and errors are not the same thing.
- Accuracy and precision have different meanings.
- Systematic errors and random errors have different causes.

Here, we look at the differences in meaning between some of the more important terms.

To do this, we will refer back to previous sections in **Topic 8D**, which involve measuring masses and volumes using different methods and apparatus.

## MISTAKES ARE NOT ERRORS

Put very simply, an error is something that even a skilled operator would find difficult to avoid, and is a consequence of the way the apparatus has been constructed and how readings can be made using it.

A mistake is something that a skilled operator can avoid by being careful.

Here are some examples of mistakes.

1 A chemist weighs a beaker on a balance without making sure the balance is tared (set to zero) beforehand. The reading on the balance could be very different from the actual mass of the beaker, so the reading should not be used, although this careless chemist may not realise this.

2 A student sees a burette reading of 27.35 $cm^3$ but writes it down as 23.75 $cm^3$. This is the student's mistake, and has nothing to do with the apparatus.

3 A student fills a burette using a funnel and forgets to remove the funnel before adding the liquid to the flask. During the addition, some of the liquid in the funnel drips into the burette, and this causes an incorrect burette reading to be recorded. This is due to the student's faulty technique. Again, it has nothing to do with the actual apparatus, only his careless use of it.

## ACCURACY AND PRECISION, AND SYSTEMATIC AND RANDOM ERRORS

The most important terms to consider are:

- **error**
- **accuracy**
- **precision**.

The last two terms are often confused. We will try to understand the difference between them by considering some burette readings and an archery competition.

A teacher and some students do a titration using the same solutions. The teacher works carefully and obtains a mean titre of 24.27 $cm^3$, which we will assume is the 'correct' value.

Consider the titres in **table A**, which were recorded by students doing the same titration as the teacher.

| STUDENT | TITRES / $cm^3$ | | | | MEAN OF ALL TITRES / $cm^3$ |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | |
| A | 24.80 | 24.85 | 24.90 | 24.80 | 24.84 |
| B | 24.95 | 24.80 | 23.25 | 23.80 | 24.20 |
| C | 24.20 | 24.30 | 24.25 | 24.20 | 24.14 |

**table A** Titres recorded by Students A, B and C doing the same titration as their teacher.

All the mean titres have been calculated correctly.

Now for the archery competition. The aim of an archery competition is to win by managing to land all the arrows as close as possible to the centre of the target. These diagrams show how well the three students have done at archery using their titre values.




**Table B** provides commentary on these values.

| STUDENT | COMMENTS |
|---|---|
| A | Student A has titres that are all concordant (within 0.10 $cm^3$ of each other), but the mean is 0.57 $cm^3$ higher than the correct value. This suggests that the titrations have been carefully carried out, but that there is probably something about the apparatus that is responsible for the large difference from the correct titre value. This is called a systematic error. The titre values are precise but not accurate. |
| B | Student B has no concordant titres (they are very different from each other) but the mean is within 0.07 $cm^3$ of the correct value. This suggests that the titrations have been carelessly done, but the student has been lucky because the mean happens to be close to the correct value. This is called a random error. Even though each individual titre is not accurate, and all four of them are not precise, the mean titre is accurate. |
| C | Student C has titres that are all concordant, and the mean is within 0.03 $cm^3$ of the correct value. This suggests that the titrations have been carefully done, and that the apparatus used is of the same standard as that used by the teacher to obtain the correct value. The titre values are both accurate and precise. |

**table B** Comments explaining how Students A, B and C have done with their titre values.

**LEARNING TIP**

Comparing the archery competition and the titrations should help you to understand the difference between accuracy and precision.

**CHECKPOINT** SKILLS CRITICAL THINKING

**1.** An experimental method requires a 25.0 $cm^3$ pipette to be used to measure a volume of liquid in different experiments. A student uses a 25 $cm^3$ measuring cylinder instead in each case. Explain whether the student has made a mistake or has introduced a random error or a systematic error.

**2.** Consider these titres recorded in $cm^3$.

| | | | | |
|---|---|---|---|---|
| Student 1 | 27.60 | 27.70 | 27.70 | 27.65 |
| Student 2 | 26.15 | 26.82 | 26.60 | 26.30 |
| Student 3 | 24.40 | 26.50 | 26.50 | 26.40 |

The teacher's mean titre was 26.50 $cm^3$, which can be assumed to be correct.

Explain whether the students' titres indicate accuracy, precision, both or neither.

**SUBJECT VOCABULARY**

**error** the difference between an experimental value and the accepted or correct value

**accuracy** a measure of how close values are to the accepted or correct value

**precision** a measure of how close values are to each other

# 8D 5 MEASUREMENT ERRORS AND MEASUREMENT UNCERTAINTIES

SPECIFICATION REFERENCE 8.22

## LEARNING OBJECTIVES

- Understand how to minimise the sources of measurement uncertainty in volumetric analysis.

## RANDOM AND SYSTEMATIC ERRORS

Consider the accepted meanings of two terms briefly mentioned in **Topic 8D.4** (**table A**).

| TERM | MEANING |
|---|---|
| **random error** | This is an error caused by unpredictable changes in conditions such as temperature or pressure, or by a difference in recording that is difficult to get exactly right. |
| **systematic error** | This is an error caused by the apparatus, and leads to the recorded value being either too low or too high. |

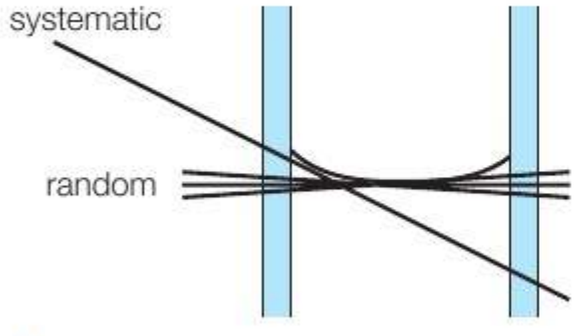
**table A**

### RANDOM ERRORS

Here are examples of random errors.

- The volume of a gas collected in a syringe is measured in different experiments done on the same day. The atmospheric pressure and the temperature of the laboratory may vary during the day, and this will cause an unpredictable change in the value recorded.
- The mass of an object is measured using the same balance but at different times during the day. The values may differ slightly, as a result of changes in temperature, draughts or condensation of water vapour on the balance pan.
- An experiment is done with a flask in an electric water bath. The thermostat is set to 50 °C, but as the heater automatically switches on, the actual temperature rises slightly above 50 °C, and then falls to slightly less than 50 °C after the thermostat switches off.

Repeating an experiment should lead to a more accurate final value being recorded because these random fluctuations become less important when values are averaged.




▲ **fig A** When recording a liquid level, the operator should view the meniscus at eye level (horizontally). This diagram shows that minor variations from the horizontal can be considered as random errors. If the meniscus is consistently viewed from above, then the error can be described as systematic.

### SYSTEMATIC ERRORS

Here are examples of systematic errors.

- A 25.0 $cm^3$ pipette has been wrongly calibrated during its manufacture, so that the graduation mark is lower than it should be. This means that no matter how carefully it is used, the volume of solution added from this pipette will always be less than 25.0 $cm^3$.
- The amount of liquid in a thermometer is more than it should be, so that the height of the liquid at all temperatures is higher than it should be. This means that the temperature recorded will always be greater than the correct temperature.
- A measuring cylinder has markings on it from 0 to 10 $cm^3$, but the diameter of the cylinder is smaller than the manufacturer intended. This means that when the liquid level shows 10 $cm^3$, the volume is less than 10 $cm^3$.

Repeating an experiment using the same apparatus will not lead to a more accurate value being recorded.

**LEARNING TIP**

Think about how using a burette could involve both random and systematic errors.

## MEASUREMENT UNCERTAINTY

When using apparatus, there is always a potential error, known as **measurement uncertainty**. The size of the measurement uncertainty is determined by the precision of the apparatus.

### BALANCES

Digital balances can have various degrees of precision. A balance that measures to three decimal places is more precise that one that measures to only one decimal place. This means that the measurement uncertainty involved in using the three-decimal place balance is lower than in using the one-decimal place balance. This is illustrated in **table B**. The greater the degree of precision of the balance, the smaller the measurement uncertainty in the recorded mass.

| NUMBER OF DECIMAL PLACES | MEASUREMENT UNCERTAINTY | EXAMPLE |
|---|---|---|
| 1 | ±0.05 g | A reading of 17.1 g could be between 17.05 and 17.15 g |
| 2 | ±0.005 g | A reading of 17.10 g could be between 17.095 g and 17.105 g |
| 3 | ±0.0005 g | A reading of 17.100 g could be between 17.0995 g and 17.1005 g |

**table B** Examples of measurement uncertainties when using a balance.




▲ **fig B** The scale on a beaker is there only as a guide and should not be used to record values to be used in calculations. A measuring cylinder is better for this purpose.

## GLASSWARE

Most laboratory glassware is manufactured to Class A and Class B (or Grade A and Grade B) standards. Class A apparatus is much more expensive than Class B apparatus, and the extra cost reflects the way in which the apparatus is calibrated, rather than the quality of the actual apparatus.

You will almost certainly use Class B apparatus, which is still capable of giving measurements to a high degree of accuracy. If you look carefully, you should be able to see the manufacturer's statement about level of precision marked on the apparatus. You should see a temperature quoted (usually 20 °C). At higher temperatures, the glass and the solution it contains will expand to different extents, and the value of the volume will no longer be accurate within the given tolerances.

**Table C** shows information about typical pieces of Class B glassware that are used to measure volumes.

| APPARATUS | CAPACITY | MEASUREMENT UNCERTAINTY |
|---|---|---|
| burette | 50 $cm^3$ | ±0.05 $cm^3$ (but the burette is read twice in a titration, so the total measurement uncertainty is ±0.10 $cm^3$) |
| pipette | 25 $cm^3$ | ±0.06 $cm^3$ |
| volumetric flask | 250 $cm^3$ | ±0.3 $cm^3$ |

**table C** Examples of measurement uncertainties in glassware.

### LEARNING TIP

Remember that, assuming you have used the piece of apparatus correctly, you still cannot say for certain that any measurement made with the apparatus is accurate. It is only possible to know if there has been an error if you happen to know the true value of the measurement made. The best you can say is that there is a potential error involved, which is given by the *measurement uncertainty* of using the apparatus.

### CHECKPOINT

SKILLS CRITICAL THINKING

1. A student carries out a titration and obtains a titre of 23.40 $cm^3$. She repeats the titration the next day using different apparatus and obtains a titre of 24.50 $cm^3$.
   (a) What is the total measurement uncertainty of the first titre?
   (b) Identify a potential source of random error that may have affected her results.
   (c) Identify a potential source of systematic error that may have affected her results.
2. A student uses a pipette and a burette to add 50.0 $cm^3$ of a liquid to a flask in different experiments. Use the table of measurement uncertainties (**table C** above) to calculate which piece of apparatus is the better one to use in this experiment.

### SUBJECT VOCABULARY

**random error** an error caused by unpredictable variations in conditions

**systematic error** an error that is constant or predictable, usually because of the apparatus used

**measurement uncertainty** the potential error involved when using a piece of apparatus to make a measurement

# 8D 6 OVERALL MEASUREMENT UNCERTAINTY

SPECIFICATION REFERENCE 8.22 PART

## LEARNING OBJECTIVES

- Understand how to estimate the overall uncertainty of the calculated result in volumetric analysis.

## PERCENTAGE MEASUREMENT UNCERTAINTIES

Each piece of apparatus you use to record a value (such as mass, volume, temperature or time) has a measurement uncertainty associated with it, which depends on the way it has been manufactured and calibrated.

The actual measurement uncertainty may be fixed. However, in many cases the **percentage uncertainty** when you use the apparatus depends on the value you measure. This mostly depends on whether you use the apparatus to record only one value or two values, and on how big the value is compared to the capacity of the apparatus.

### GLASSWARE

**Table A** shows typical percentage uncertainties for common items of glassware.

| APPARATUS | CAPACITY | UNCERTAINTY | PERCENTAGE UNCERTAINTY |
|---|---|---|---|
| burette | 50 $cm^3$ | ±0.05 $cm^3$ | Note that two burette values are read in a titration, so the total measurement uncertainty is ±0.10 $cm^3$. If the titre is 22.50 $cm^3$, then the percentage uncertainty is: ±0.10 × 100/22.50 = ±0.44% |
| pipette | 25 $cm^3$ | ±0.06 $cm^3$ | The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.06 × 100/25 = ±0.24% |
| volumetric flask | 250 $cm^3$ | ±0.3 $cm^3$ | The reading is taken only once, and for the same volume each time, so the percentage uncertainty is always: ±0.3 × 100/250 = ±0.12% |

**table A** Typical percentage uncertainties for a burette, pipette and volumetric flask.

### BALANCES

The percentage uncertainty in using a balance depends on:

- the precision of the balance, i.e. the number of decimal places to which the balance can be read
- the mass being weighed, as the percentage uncertainty will be greater for a smaller mass.

#### WORKED EXAMPLE 1

- mass of a marble chip = 3.57 g

The measurement uncertainty in a two-decimal place balance is ±0.005 g. We have to count this uncertainty twice. This is because there is a ±0.005 g uncertainty when calibrating the balance to zero, as well as a ±0.005 g uncertainty when measuring the mass.

So the percentage uncertainty is:

$$\pm\frac{2 \times 0.005 \times 100}{3.57} = \pm 0.28\%$$

## WORKED EXAMPLE 2

- mass of weighing bottle + solid = 20.354 g
- mass of weighing bottle = 19.816 g
- mass of solid = 0.538 g

The measurement uncertainty in a three-decimal place balance is ±0.0005 g. We have to count this uncertainty twice for each measurement and we have two measurements, so we need to count it four times. The percentage uncertainty is:

$$\pm\frac{4 \times 0.0005 \times 100}{0.538} = \pm 0.37\%$$

The percentage uncertainty is greater in Worked example 2, even though the balance reads to one more decimal place. This is because the balance is used twice and also because the mass being weighed is much smaller.

### ADDING MEASUREMENT UNCERTAINTIES

If a final answer has been obtained using more than one piece of apparatus, then the approximate total measurement uncertainty is obtained by adding together the individual uncertainties.

For example, if a concentration of 0.118 mol dm$^{-3}$ has been calculated from a titration that has involved balance and glassware uncertainties, then the uncertainties might be:

| | |
|---|---|
| balance | ±0.09% |
| volumetric flask | ±0.12% |
| pipette | ±0.24% |
| burette | ±0.47% |
| overall percentage uncertainty | ±0.92% |

This means that there is an overall uncertainty in the concentration of 0.92% of 0.118, which is about ±0.001, so the final value can be quoted like this:

concentration = 0.118 ± 0.001 mol dm$^{-3}$

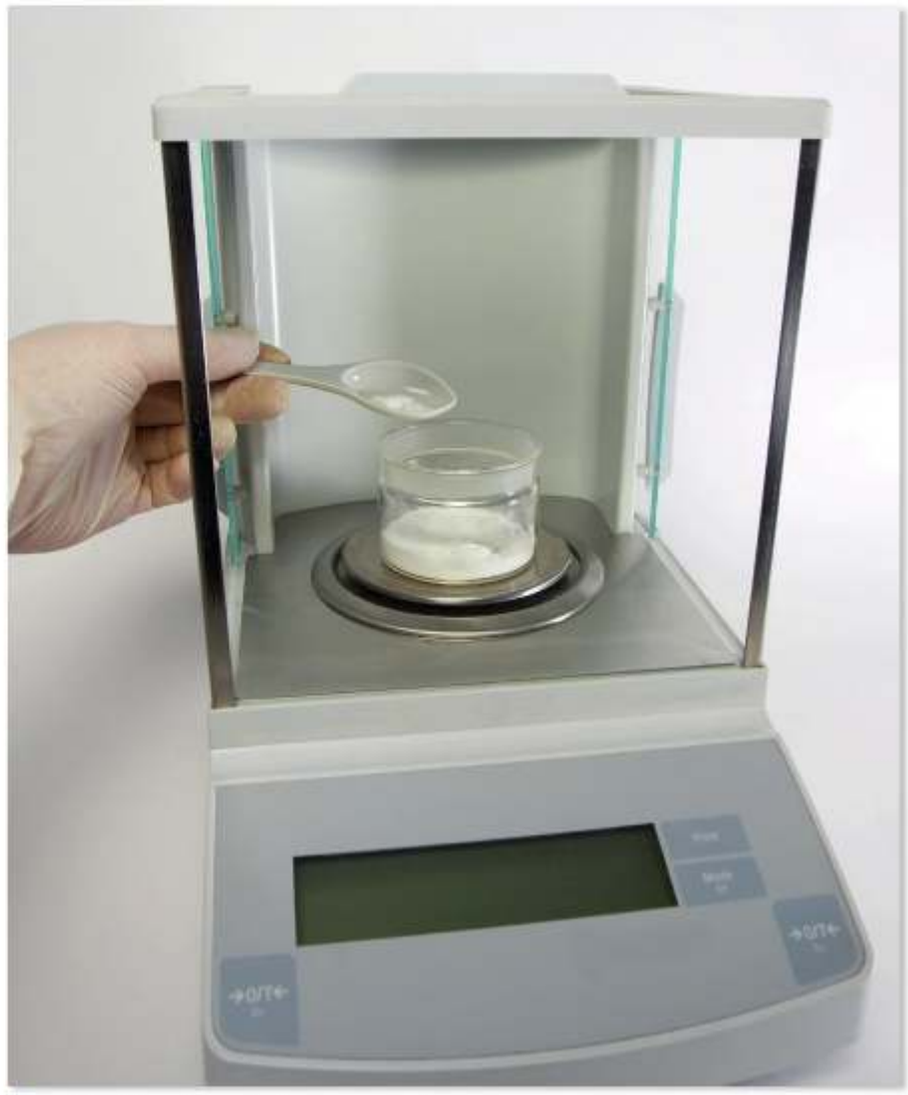
which means that the exact value is in the range 0.117–0.119 mol dm$^{-3}$.

### MINIMISING ERROR AND UNCERTAINTY

How can errors and uncertainties be minimised in an experiment? This depends on a number of factors, some of which are easier to control than others.

For example, in a thermochemistry experiment carried out in the laboratory using standard equipment, there will always be transfer of heat energy to the surroundings. This heat transfer creates a random error in the measurement of the temperature change. In this type of experiment, using a balance that reads to three decimal places (instead of a balance that reads to two decimal places) will have no significant effect on the overall uncertainty of the final value. Minimising heat energy losses will have a much greater effect.

Where a measuring instrument can be used for a range of values, the percentage uncertainty can be minimised by using a higher value rather than a lower one. For example, when using a balance, weighing a sample of 5 g will lead to a much lower percentage uncertainty than weighing a 0.5 g sample. In a titration, it is not a good idea to have a titre value of 10 cm$^3$ instead of 30 cm$^3$, as the larger titre has the lower percentage uncertainty.



▲ **fig A** This balance is tared (set to zero) and ready to use. The pan is enclosed; this is necessary on a four-decimal place balance because of fluctuations in the reading caused by draughts of air.

### LEARNING TIP

Practise calculating measurement uncertainties and percentage uncertainties for different pieces of apparatus.

### CHECKPOINT

1. A student uses a one-decimal place balance to weigh a piece of zinc. He records a mass of 2.8 g. What is the percentage uncertainty?
2. A thermometer has a measurement uncertainty of 1 °C. A student uses it to measure a temperature rise and records these values:

   start temperature = 15 °C

   final temperature = 28 °C

   What is the percentage uncertainty in the temperature rise?

### SUBJECT VOCABULARY

**percentage uncertainty** the actual measurement uncertainty in an experiment multiplied by 100 and divided by the value recorded